

Sampling

Nyquist Sampling Theorem:

$$f_s > 2 \cdot f_{\text{dynamics}}$$

Sampling time

$$T_s = \frac{\tau}{20}$$

, τ dominant pole time constant
→ how long it takes for robot to fall over without any control action

find how long it takes from small initial perturbation until at 90° . $\Rightarrow 0.75$

$$T_s = \frac{0.7}{20} = 0.035 = 35 \text{ ms}$$

$$f_s = \frac{1}{0.035} = 28.57 \text{ Hz} \quad \text{AT LEAST}$$

Do 200 Hz to be safe